

FIDELITY-PRESERVING PERCEPTUAL IMAGE COMPRESSION VIA A RATE-AWARE MIXTURE OF LORA EXPERTS

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ABSTRACT

Learned image compression (LIC) is typically optimized for rate–distortion performance, yet perceptual fine-tuning often shifts the codec away from its original fidelity-oriented operating point. To address this, we propose a parameter-efficient framework that decouples perceptual adaptation parameters from the fidelity-oriented parameters. Specifically, we insert low-rank adaptation (LoRA) modules into the synthesis transform and fine-tune them with perceptual objectives. By simply disabling these updates, the original decoder is recovered exactly, enabling reversible switching between fidelity and perception modes for the same bitstream. Furthermore, to address rate-dependent artifacts, we introduce a rate-aware mixture of LoRA experts, in which a soft router adaptively combines specialized low-rank updates conditioned on the quality level. Experiments show that the proposed method preserves the original rate–distortion performance while achieving superior perceptual quality compared to state-of-the-art approaches, with only 11.62 M additional parameters¹.

Index Terms— Fidelity and Perception, Mixture of LoRA Experts, Learned Image Compression

1. INTRODUCTION

Learned image compression (LIC) has emerged as a promising alternative to conventional codecs, achieving competitive rate–distortion performance through jointly optimized non-linear transforms and probability models [1]. Yet the rate–distortion objective is only half of the story: how a codec spends its limited bit budget is dictated by its reconstruction objective, and this choice has profound consequences for what the reconstructed image looks like.

MSE-optimized codecs produce faithful reconstructions given sufficient bits, but yield blurred and over-smoothed images at low bitrates. Their objective pulls the reconstruction toward the source, prioritizing fidelity at the expense of realism (Fig. 1(a)). Perceptual compression reverses this trade:

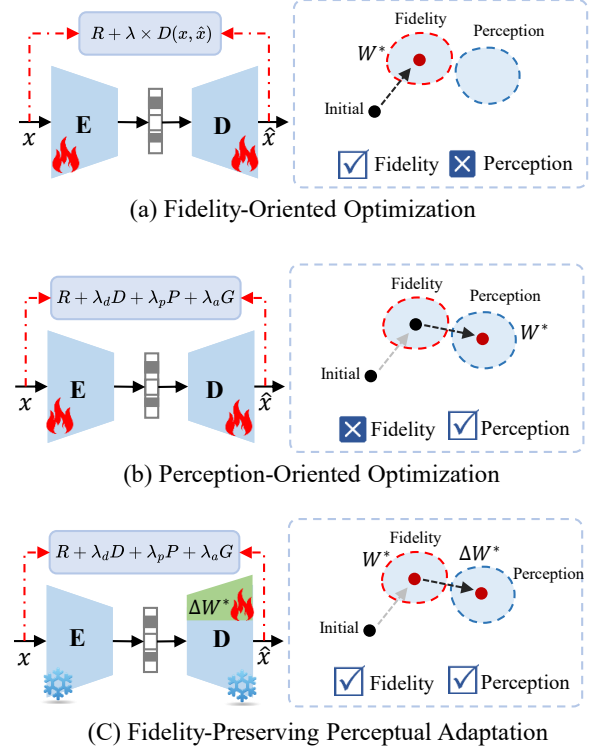


Fig. 1. Optimization process of image compression models with different objectives. (a) Fidelity objective. (b) Perception objective. (c) Fidelity-preserving perceptual adaptation.

by incorporating feature-based losses and generative priors—e.g., adversarial training in HiFiC [2] and conditional diffusion decoders that synthesize texture from transmitted latents [3, 4, 5]—these methods improve perceptual quality at the cost of pixel-level fidelity (Fig. 1(b)).

This tension is not an artifact of specific architectures but a fundamental limit: the rate–distortion–perception tradeoff dictates that, among optimal codes, improving perceptual realism at fixed bitrate necessarily increases distortion [6]. A natural remedy is therefore to expose multiple operating points from a single representation. Agustsson et al. [7] enable the receiver to adjust the distortion–realism balance

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¹ Code is released at <https://github.com/cx-333/fppa>

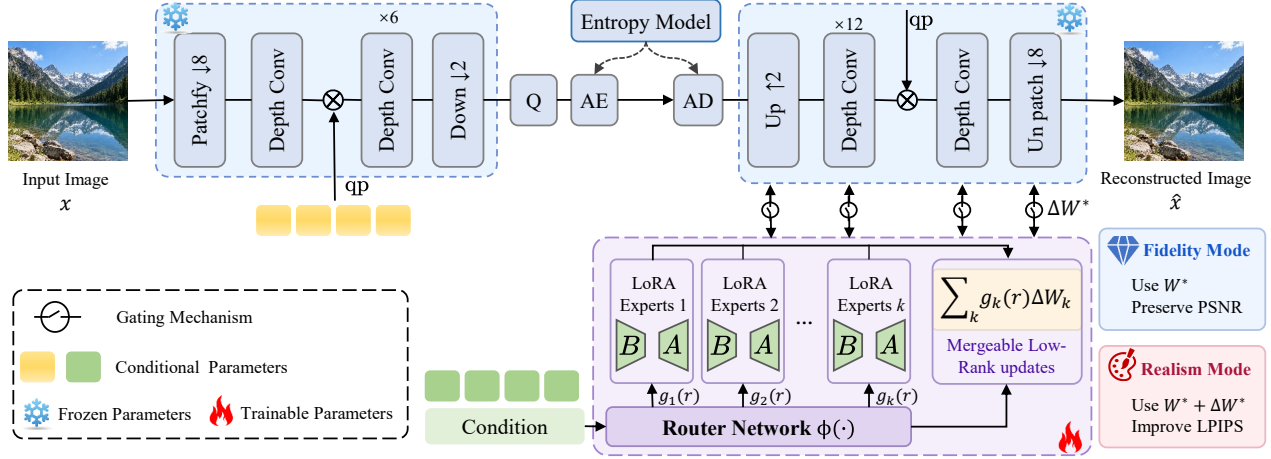


Fig. 2. Overview of the proposed framework. The mixture of LoRA experts learns rate-aware perceptual delta updates that are decoupled from the original fidelity-optimal solution, preserving fidelity while driving the model toward the perceptual optimum.

via a conditional generator, and Iwai et al. [8] extend this to variable-rate compression. These approaches, however, suffer from two drawbacks. First, they attach full-capacity generators trained jointly with perceptual objectives, whose size dominates the storage budget of the codec. Second, because the decoder itself is modified, the pretrained fidelity-oriented operating point is lost and cannot be recovered once the perceptual mode is activated.

To overcome these limitations, we propose a parameter-efficient framework that disentangles perceptual adaptation from the pretrained parameters responsible for reconstruction fidelity (Fig. 1(c)). Specifically, we freeze the entire codec and optimize only compact low-rank branches injected into a handful of decoder layers. This design incurs negligible additional storage, leaves the codec—and hence the decoded bitstream—completely intact, and enables on-the-fly mode switching by simply exchanging the lightweight adapter weights. The contributions are summarized as follows:

- **Fidelity-preserving adaptation.** We propose a parameter-efficient framework adapting pretrained codecs for perceptual image compression via lightweight low-rank decoder branches, leaving the transmitted representation, entropy model, and rate-distortion operating point intact by construction.
- **Bitstream-reversible parameterization.** We propose a low-rank update rule letting one coded bitstream serve both decoding modes: activation yields perceptually enhanced reconstructions, while bypassing provably degenerates to the original decoder for fidelity-oriented reconstruction without re-encoding.
- **Rate-aware LoRA expert mixture.** We propose a

mixture of LoRA experts conditioning low-rank updates on the rate/quality operating point, with only 11.6 M additional parameters, versus 35.9 M for a per-quality LoRA.

2. METHODOLOGY

2.1. Overview

Fig. 2 illustrates the proposed framework. The pretrained codec remains frozen; only low-rank adapter branches in decoder layers are trainable. Given quality index q , a router mixes adapter updates from a learned vector bank. Disabling the adapters (fidelity mode) reproduces the original rate-distortion operating point, whereas enabling them (perceptioin mode) applies the mixed update ΔW to the same bitstream.

Let x be an input image and $q \in \{0, 1, \dots, q_{\max}\}$ a quality index. The codec comprises an analysis transform $E(\cdot)$, quantization $Q(\cdot)$, an entropy model, and a synthesis transform $G(\cdot)$.

$$\hat{y}_q = Q(E(x, q; \theta_E)), \quad \hat{x}_0 = G(\hat{y}_q, q; \theta_G). \quad (1)$$

Rate-distortion training optimizes:

$$L_{RD}(\Theta) = \mathbb{E}_{(x, q) \sim \mathcal{D}} [R(x, q; \Theta) + \lambda_D(q) D(x, \hat{x}_0)], \quad (2)$$

where Θ collects the codec parameters, $\mathcal{D}$ is the data distribution, R measures expected bits per pixel including side information, and D is a distortion metric such as MSE. We call Θ_0 in Eq. (2) the fidelity-oriented solution, without a global-optimum assumption.

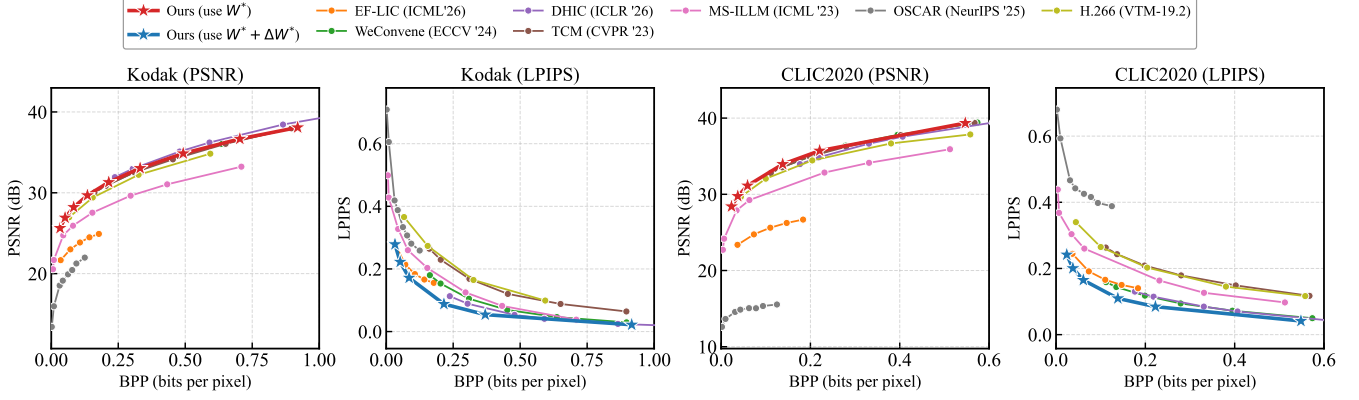


Fig. 3. Comparisons of rate-distortion and rate-perception curves across different methods on the Kodak and CLIC2020 dataset.

Perceptual finetuning, which augments Eq. (2) with perceptual and adversarial losses, generally moves all parameters away from Θ_0 , forfeiting the pretrained operating point, whereas perceptual adaptation should leave Θ_0 independently recoverable. We therefore freeze the codec and learn perceptual corrections only through lightweight additive branches.

2.2. Fidelity-Preserving Low-Rank Adaptation

For a frozen layer weight $\mathbf{W}_0 \in \mathbb{R}^{d_o \times d_i}$, we attach a low-rank additive update [9]:

$$\Delta \mathbf{W} = s \mathbf{B} \mathbf{A}, \quad \mathbf{W}_P = \mathbf{W}_0 + \Delta \mathbf{W}, \quad s = \alpha / r, \quad (3)$$

where $\mathbf{A} \in \mathbb{R}^{r \times d_i}$, $\mathbf{B} \in \mathbb{R}^{d_o \times r}$, r is the base rank, and α is a fixed scaling constant.

For a $k \times k$ convolution (C_i input channels, C_o output channels, g groups), we use ConvLoRA factors of shapes $(rk, C_i k)$ and $(C_o k / g, rk)$, giving effective update rank rk , and each expert adds $rk^2(C_i + C_o / g)$ parameters. The product is reshaped into $\mathbf{W}_0$, so the update modifies the kernel itself rather than inserting two spatial convolutions.

Let $m \in \{0, 1\}$ select the reconstruction mode. The effective weight is

$$\mathbf{W}(m) = \mathbf{W}_0 + m \Delta \mathbf{W}, \quad (4)$$

with the same switch for every bias increment. Setting $m = 0$ exactly recovers the original synthesis transform, provided the pretrained parameters and inference state (including running statistics) remain untouched, thus guaranteeing preservation of the base mode rather than identical distortion in the adapted mode.

2.3. Rate-Aware Mixture of LoRA Experts

As perceptual corrections vary with bitrate, a single update shared across quality levels is suboptimal (see Table 1). We

therefore assign K experts per adapted layer, combined by a router conditioned on q . For layer ℓ , a learnable embedding table $\mathbf{T}_\ell \in \mathbb{R}^{(q_{\max}+1) \times 128}$ provides the router input by indexing:

$$\mathbf{e}_{\ell,q} = \mathbf{T}_\ell[q], \quad \mathbf{a}_\ell(q) = \mathbf{U}_\ell \mathbf{e}_{\ell,q} + \mathbf{b}_\ell, \quad (5)$$

with $\mathbf{U}_\ell \in \mathbb{R}^{K \times 128}$ and $\mathbf{b}_\ell \in \mathbb{R}^K$ producing K expert logits. Softmax gating yields coefficients

$$\pi_{\ell,j}(q) = \frac{\exp(a_{\ell,j}(q))}{\sum_{i=1}^K \exp(a_{\ell,i}(q))}, \quad j = 1, \dots, K, \quad (6)$$

and the effective weight of layer ℓ is

$$\mathbf{W}_\ell(q, m) = \mathbf{W}_{\ell,0} + m s_\ell \sum_{j=1}^K \pi_{\ell,j}(q) \text{Reshape}(\mathbf{B}_{\ell,j} \mathbf{A}_{\ell,j}), \quad (7)$$

where $s_\ell = \alpha_\ell / r_\ell$ is the per-layer scaling. Experts use independently initialized $\mathbf{A}_{\ell,j}$ and zero-initialized $\mathbf{B}_{\ell,j}$, breaking expert symmetry while preserving the initial base reconstruction. With $\pi_{\ell,j}$ depending only on q , the mixed update is assembled once per quality level and cached as one convolutional kernel. The quality index q is a rate-control condition rather than realized bits per pixel, so expert specialization is learned from data, independent of fixed rate intervals.

3. EXPERIMENTS

3.1. Experimental Setting

Following [10], all experiments in this paper are conducted on images selected from ImageNet [11], COCO 2017 [12], DIV2K [13], and Flickr2K [14], which together constitute a training set of approximately 10^5 images². During fine-tuning, the images are cropped into 512×512 patches. The

²https://github.com/JiangWeibeta/MLIC/blob/main/train_list.txt



Fig. 4. Visual comparison of cropped patches at similar bitrates. LPIPS and DISTS are reported below each image (lower is better).

rate-perception optimization is performed for 10^4 steps with a learning rate of 1×10^{-4} .

For evaluation, we use the Kodak [15] and CLIC2020 [16] datasets as test sets. We compare our method against both conventional codecs, including H.266/VVC [17], and state-of-the-art learned image codecs [18, 19, 20, 21].

3.2. Performance Evaluation

3.2.1. Rate-Distortion-Perception Performance

Fig. 3 compares the rate-distortion and rate-perception curves of different methods across bit rates on the Kodak and CLIC2020 datasets. The proposed method is evaluated in two settings: the fidelity mode, where the adaptation branch is disabled (W^*), and the perception mode, where it is enabled ($W^* + \Delta W^*$). In the fidelity mode, the proposed method outperforms all comparison methods except DHIC [19] in terms of PSNR, whereas in the perception mode, it outperforms all comparison methods in terms of LPIPS. Note that the reconstructions of DHIC exhibit poor perception.

3.2.2. Qualitative Comparisons

Fig. 4 compares cropped patches of a motorcycle image at similar bitrates (0.072–0.085 bpp), where the marked region

contains dense high-frequency structures, including wheel spokes, metal rims, and guardrail edges. H.266 smooths the spoke structure heavily and loses most fine details, yielding the highest LPIPS (0.30) among all methods. MS-ILLM and EF-LIC recover the overall wheel layout, but their reconstructions still show blurred spokes and softened metal edges. OSCAR produces noticeable texture artifacts, and its LPIPS (0.49) is the worst here. At a lower bitrate (0.072 bpp), the proposed method preserves the sharpness of the spokes and rim edge clarity while keeping natural color and contrast. It attains the best LPIPS (0.16) and a competitive DISTS (0.08) at the lowest bitrate, indicating that the fidelity-preserving adaptation recovers perceptually important details that conventional codecs and existing perceptual methods miss.

3.3. Ablation Study

Table 1. Ablation Study on Component Effectiveness

Method	Params (45.71 + ΔW)	BD-Rate↓	
		LPIPS	DISTS
Single LoRA	4.49 M	50.50 %	40.69 %
Multi-LoRA	4.49×8 M	-1.54 %	-1.88 %
Ours	<u>11.62</u> M	<u>0.00</u>	<u>0.00</u>

Table 1 ablates the rate-aware mixture of LoRA experts. A single LoRA shared across all quality levels performs substantially worse than the full model (50.50% and 40.69% BD-rate in LPIPS and DISTS), confirming that one update direction cannot accommodate the perceptual corrections required at different bitrates. Multi-LoRA, storing an independent expert per quality level, only marginally outperforms our mixture while requiring 4.49×8 M adapter parameters, about 3 times the 11.62 M of our design. These results indicate that rate-aware gating of shared low-rank experts achieves rate-adaptive perceptual compression at a small fraction of the storage cost.

4. CONCLUSION

In this work, we present a parameter-efficient framework for perceptual adaptation in learned image compression. Through low-rank adaptation of the synthesis transform, the framework preserves the fidelity-optimal solution and decouples perceptual optimization from the fidelity-oriented parameters. Moreover, we introduce a rate-aware mixture of LoRA experts to support multi-rate coding, in which a soft router combines specialized low-rank updates according to the quality level. Experiments demonstrate that the proposed method preserves the original rate-distortion performance while improving perceptual quality and introducing only a small number of additional trainable parameters.

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